

FAULT CODE 111 - Engine Control Module Critical Internal Failure - Bad Intelligent Device or Component

Warnings and Cautions

⚠ CAUTION ⚠

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

Troubleshooting Summary		
STEPS		SPECIFICATIONS
STEP 1.	Check the ECM.	
	STEP 1A. Check the ECM.	Fault Code 111 inactive?
	STEP 1B. Check the inactive counts of Fault Code 111.	Less than 3 counts?
STEP 2.	Check ECM calibration and clear the fault codes.	
	STEP 2A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 2B. Disable the fault code.	Fault code inactive?

STEP 1. Check the ECM.

STEP 1A. Check the ECM.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check the ECM. • Turn the keyswitch OFF and wait 5 seconds. • Start the engine and let it idle for 1 minute.	Fault Code 111 inactive? YES	1B
	Fault Code 111 inactive? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	2A

STEP 1B. Check the inactive counts of Fault Code 111.**Conditions :**

- Turn the keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check the inactive counts of Fault Code 111. • Use INSITE™ electronic service tool to read the inactive counts of Fault Code 111.	Less than 3 counts? YES	2B
	Less than 3 counts? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	Repair complete.

STEP 2. Check ECM calibration and clear fault codes.**STEP 2A. Check if an ECM calibration update is available.**

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. • Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Dataplate in Features and Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	2B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)	2B

STEP 2B. Disable the fault code.**Conditions :**

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault code inactive? YES	Repair complete.
	Fault code inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	QSK50 CM2150 MCRS	4022102

Last Modified: 28-Jul-2016